

**Project Design Phase-II  
Technology Stack (Architecture & Stack)**

|               |                                                                                             |
|---------------|---------------------------------------------------------------------------------------------|
| Date          | 30 June 2026                                                                                |
| Team ID'S     | SWUID2026211879<br>SWUID2026196379<br>SWUID2026211883<br>SWUID2026211881<br>SWUID2026226496 |
| Project Name  | Smart Lender                                                                                |
| Maximum Marks | 4 Marks                                                                                     |

**Technical Architecture:**

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

**Example: Order processing during pandemics for offline mode**

**Reference:** <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>

**References:**

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>

**Table-1 : Components & Technologies:**

| S.No | Component                       | Description                                        | Technology                              |
|------|---------------------------------|----------------------------------------------------|-----------------------------------------|
| 1.   | User Interface                  | Web interface for loan application and prediction. | HTML, CSS, JavaScript, React.js         |
| 2.   | Application Logic-1             | User registration, login and authentication.       | Python (Flask)                          |
| 3.   | Application Logic-2             | Loan application processing and validation.        | Python, Pandas                          |
| 4.   | Application Logic-3             | AI-based loan approval prediction.                 | Scikit-learn, NumPy                     |
| 5.   | Database                        | Store user, application and prediction details.    | MySQL                                   |
| 6.   | Cloud Database                  | Cloud-based database service.                      | Firebase / MySQL Cloud                  |
| 7.   | File Storage                    | Store uploaded documents and reports.              | Local File System / Firebase Storage    |
| 8.   | External API-1                  | Email notification service.                        | SMTP / Email API                        |
| 9.   | External API-2                  | OTP verification service.                          | Twilio OTP API                          |
| 10.  | Machine Learning Model          | Loan Approval Prediction Model.                    | Random Forest Classifier (Scikit-learn) |
| 11.  | Infrastructure (Server / Cloud) | Application deployment on server / cloud.          | Local Server / Render / Railway / AWS   |

**Table-2: Application Characteristics:**

| S.No | Characteristics          | Description                                                                                                               | Technology                        |
|------|--------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| 1.   | Open-Source Frameworks   | List the open-source frameworks used.                                                                                     | React.js, Flask, Scikit-learn     |
| 2.   | Security Implementations | List all the security / access controls implemented, use of firewalls etc.                                                | SHA-256, HTTPS                    |
| 3.   | Scalable Architecture    | Justify the scalability of architecture (3 – tier, Micro-services).                                                       | Three-Tier Architecture           |
| 4.   | Availability             | Justify the availability of application (e.g. use of load balancers, distributed servers etc.).                           | Cloud Hosting                     |
| 5.   | Performance              | Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDN's etc). | Pandas, NumPy, Optimized ML Model |